

How neural meets symbolic

A simple guide to this lab (and the online demo)

Short companion to the full progress notes — for readers new to neuro-symbolic AI.

Demo: <https://udonsikalu.github.io/cxr-evidence-grounding-lab/>

Full record: [notes/progress-notes.pdf](#)

Read this if you want the idea in plain language, then open the demo.

Read the full notes if you want every protocol number, held-out table, and freeze date.

1 The idea in one picture

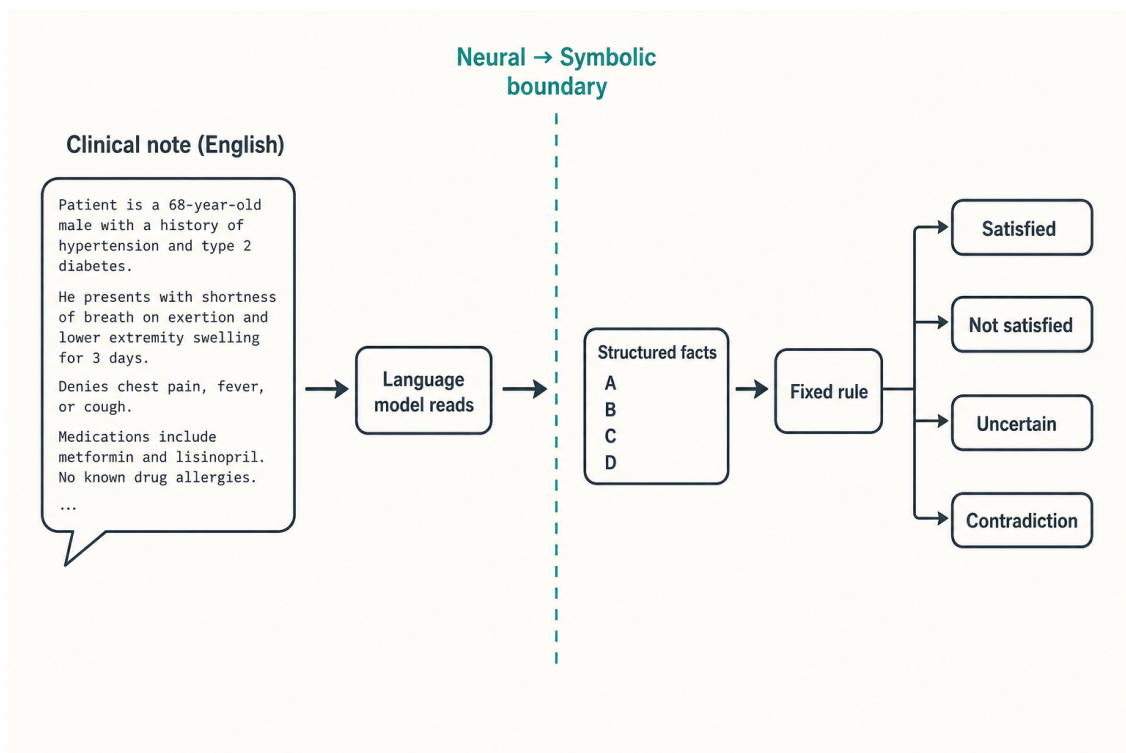


Figure 1: A language model reads the note. Structured facts cross a boundary into a fixed rule. The rule — not the model — issues the final label.

Neuro-symbolic means: neural (the model) + symbolic (clear rules). The interesting failure is not “the model is dumb.” It is: *the model seemed to understand, but the formal structure dropped the point.*

2 Our toy question

One made-up clinical question (synthetic notes, not real patients):

Did **first-line therapy** fail?

A fixed rule needs four facts (and a contradiction flag):

Atom	Plain meaning
A	A first-line therapy is named
B	It was actually given / tried
C	Something counts as failure
D	That failure is of <i>that</i> first-line regimen
X	The note contradicts itself about the above

Same atoms in $\rightarrow$ same verdict out: Satisfied, Not satisfied, Uncertain, or Contradiction.

3 Two ways to answer the same note

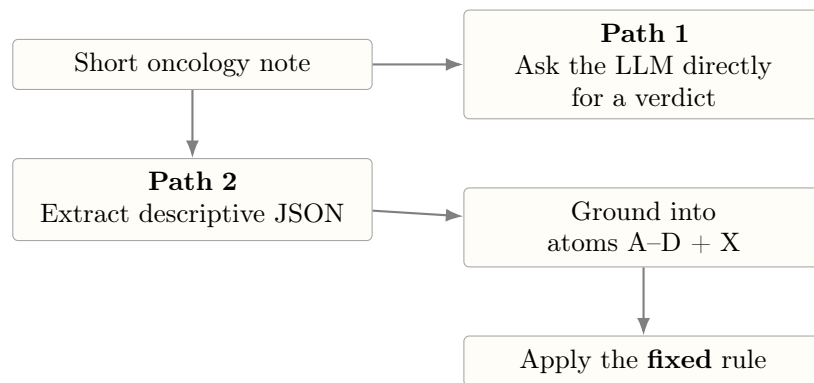


Figure 2: We always compare both paths on the same note. Gold labels are never shown to the model.

If Path 1 and Path 2 disagree, something was lost at the boundary.

4 What the online demo shows

The public page is **replay only**: it loads frozen experiment files. It does **not** run a model on anyone's computer.

N2S evidence grounding — findings demo
Artifact replay only — frozen Phase 1–4 results. No live models, no calls to anyone's GPU. Clone the repo to run Ollama locally on :8253 / :8254 .

ARTIFACT: Phase-4 held-out (C9-U12) MODEL: llama3.8b-instruct-q4_0 **Load** loaded phase4-heldout-panel.json · replay only (no inference)

Model: llama3.8b-instruct-q4_0 A 2/7 B 4/7 C 6/7 D 5/7 D rep-loss misses: 1

llama3.8b-instruct-q4_0 A2/7 mistral:instruct A7/7 qwen2.5-coder:32b A2/7

Case	Verdict	Match	Loss stage
A	NOT SATISFIED	no	direct_verdict_miss
B	UNCERTAIN	no	analysis_output_misalignment
C	CONTRADICTION	yes	none
D	CONTRADICTION	yes	none

C9 — GOLD CONTRADICTION - CONTRADICTION
 07:40 tumor-board list: first-line ipilimumab/nivolumab has failed; schedule second-line BRAF-directed therapy. 19:10 same-day clinic addendum: restaging today is stable disease on first-line ipilimumab/nivolumab; keep current immunotherapy, do not switch.
 Same-day reverse: morning note says first-line failed and switch; evening note says continue first-line with stability.

A — DIRECT VERDICT
 verdict: NOT SATISFIED ✗ gold
 loss: direct_verdict_miss
 The note states that the first-line therapy has failed and a second-line treatment is scheduled, but then it contradicts this by saying to continue the current immunotherapy without switching.

B — ANALYSIS - VERDICT
 verdict: UNCERTAIN ✗ gold
 loss: analysis_output_misalignment
 The note suggests that a first-line therapy is identified (ipilimumab/nivolumab), as explicitly stated in the tumor-board list at 07:40. This meets condition A.
 Regarding condition B, it appears that the first-line therapy was given or attempted, as the note mentions "first-line ipilimumab/nivolumab has failed", implying that the treatment was administered before its failure.
 The note also indicates a failure event (condition C) with the statement "first-line ipilimumab/nivolumab has failed" in the tumor-board list. This suggests that the patient's disease progressed or did not respond to the first-line therapy, meeting condition C.
 However, the addendum at 19:10 contradicts this notion, stating that restaging shows stable disease on the same first-line ipilimumab/nivolumab regimen. This implies that the failure event may not be a true progression or non-response, but rather a

Figure 3: GitHub Pages demo — pick an artifact and a model, then click cases in the left list. Live URL: <https://udonsikalu.github.io/cxr-evidence-grounding-lab/>

4.1 H

ow to walk the demo (2 minutes)

1. Open the link above (or open docs/ locally).
2. Keep the default **Phase-4** artifact, or try Phase-2 / Phase-3.
3. Click the model buttons (Llama / Mistral / Qwen) — same notes, different models.
4. Click a case on the left. Read the note, then the A–D table.

4.2 W

hat A–D mean in the demo

These are **not** four medical rules. They are four places meaning can disappear:

Col.	Plain meaning
A	Direct LLM verdict (no structure).
B	Model writes an analysis, then a final answer — does the answer keep what the analysis said?
C	Full pipeline: extract → atoms → fixed rule.
D	Analysis had the distinction, but the structured form dropped it (“representation loss”).

Loss stage under each column tells you *where* it broke, not just that it missed gold.

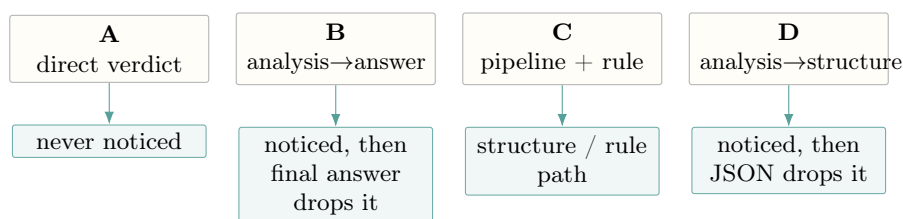


Figure 4: Conditions A–D map to different failure stories. That is why the demo shows all four.

5 The story in three beats

1. **Make the boundary visible.** Same notes, direct LLM vs structured pipeline (Milestone 1 UI on port 8253 if you run locally).
2. **Keep hard states on purpose.** Mark contradiction and uncertainty in the structured form so the rule can fire — states the raw LLM often narrates and then throws away.
3. **Ask where meaning died.** Conditions A–D on discovery notes, then held-out notes, across three local models. Representation loss (D) still appeared on unseen wording.
4. **Verify, then abstain safely (Phase 5).** If structured form may not match what the analysis said, regenerate once; if still unverifiable, emit **Review** — not a guessed verdict, and not the same as clinical Uncertain.
5. **Stack more checks (Phase 6–7).** Round-trip + dual-path, then an evidence span gate; try the same idea on BMT/CAR-T wording. Safer auto-labels often mean more Reviews.
6. **Rep-eng pilot (Phase 9–10).** On one BMT false-contradiction case: localize the repair commit, then show hidden activation steering (not logit bias) can flip the model’s contradiction choice while controls hold.

That is the arc. The long PDF lists every count and protocol freeze.

6 Phase 5 in one sentence

When the model’s JSON might have dropped what its own analysis noticed, the system checks, tries once to repair, then either applies the fixed rule or says **Review**. On the Phase 4 held-out slice, every measured D “representation loss” was either fixed or sent to Review — not left as a silent wrong label.

7 Phase 6–7 in one sentence

Phase 6: if two paths disagree or a paraphrase of the structure does not match the note, Review. Phase 7: every formal claim must carry a source span; same stack on six BMT/CAR-T notes — Llama/Mistral Dual_full safety 1.0 (lower coverage); Qwen coder 0.75; evidence gate barely moved the needle. Soft claim only.

8 Phase 8 in one sentence

A mid Qwen 2.5 14B and a non-coder Qwen 2.5 32B were run on the same safety stack. On BMT/CAR-T, non-coder 32B Dual safety reached 1.0 (vs frozen coder 0.75); 14B still missed once. Size helps some; protocol still required. Soft claim only.

9 Phase 9 in plain language (rep-eng pilot, frozen Aug 2026)

Problem case BC_E1: a note describes response then progression (normal course). Gold = Satisfied. Some models still mark **contradiction.present** = **true** after repair and can AUTO-label Contradiction — wrong.

What we did (Observe → Localize → Intervene):

- **9A:** Same note + schema on HuggingFace Qwen 7B vs 14B. 7B repair sets X=true; 14B stays false.
- **9B:** At the repair step where the model writes **present:** **true/false**, 7B logit margin favored true (+1.6); 14B favored false (−4.0).
- **9C:** We added +4 logit bias toward **false** *only at that commit*. BC_E1 flipped to X=false and verdict Satisfied. Two control cases unchanged.

What that proves: the error is causally tied to the **repair boolean commit**, not immovable.

What Phase 9 alone did *not* prove: hidden-representation steering (9C used logit bias).

Phase 10 (below) addressed that.

10 Phase 10 (frozen Aug 2026 — genuine pilot rep-eng)

Stricter question answered: yes — on BC_E1, patching hidden layers at the repair commit (no logit bias) made 7B *naturally* commit **false**.

Method: fixed direction = BC_E2 false-commit hidden minus BC_E1 true-commit hidden, layers 0.75 + 1.00; patch only at **contradiction.present** commit.

Result @ $\alpha=4$: logit margin +1.6 → −0.1; X true→false; verdict Contradiction→Satisfied. Controls BC_C1 / BC_E2 unchanged.

Claim (pilot): contrast-derived hidden patch **causally changed** classification on BC_E1 while preserving two controls.

Do not overclaim: we found a *steerable direction*, not “the contradiction representation.” One case; α tuned on target.

Next: Phase 11 — does the **same frozen vector** generalize?

11 Phase 11 (frozen Aug 2026 — partial generalization)

Question: same Ph10 vector + $\alpha=4$ on **unseen** held-out notes?

Result: gate **NO** but informative — controls 3/3 held; BC11_E2 fixed; BC11_E3 still wrong (margin dropped only).

Reading: direction generalizes to at least one unseen course case; not universal. Freeze as soft partial result.

Artifact: artifacts/phase11-generalization-panel.json. Explorer: <http://127.0.0.1:8256/> (cxr-evidence-grounding-lab-repeng/).

12 Phase 12A (frozen Aug 2026 — margin / strength regime)

Question: Ph11’s miss on E3 — wrong direction, or same direction but not strong enough?

Method: same frozen Ph10 vector; $\alpha \in \{2, 4, 6, 8\}$ on BC11_E2 and BC11_E3; controls stay at $\alpha=4$.

Result: mechanism gate **YES**. Margins move monotonically toward **false**. E2 flips at $\alpha=4$ (baseline margin +1.13). E3 flips at $\alpha=8$ (baseline +2.38).

Reading: Ph11 and Ph12 are complementary. Fixed $\alpha=4$ was too weak for E3; the direction still works. Soft claim on these two cases only — not “all failures are margin-regime.” Freeze before adaptive- α experiments.

Artifact: `artifacts/phase12-margin-regime-panel.json`. Protocol: `docs/PHASE12-PROTOCOL.md`.

13 Phase 12B (frozen Aug 2026 — second held-out; gate NO)

Question: does Ph12A's LOW/HIGH α rule transfer to **new** notes?

Result: gate **NO**. All three targets HIGH; margins moved toward false but none flipped by $\alpha=8$. Contradiction control BC12_C1 broken at $\alpha=4$ (true→false). N1 held.

Reading: direction still has leverage; "HIGH $\Rightarrow$ flip @8" did not transfer; more strength is not free. Soft negative — freeze before adaptive- α .

Artifact: `artifacts/phase12b-heldout-panel.json`. Protocol: `docs/PHASE12B-PROTOCOL.md`.

14 Phase 13 (frozen Aug 2026 — representation geometry; gate YES)

Question: at the repair commit (no steer), do course false-X notes and true-contradiction notes sit in separable regions along the Ph10 direction?

Result: gate **YES** (soft). 5/5 course + 4/4 contradiction usable. $\cos@1.00$ ranges do not overlap (gap ≈ 0.021). Leave-one-out probe accuracy 0.889.

Reading: classes look softly separable — useful for designing safer conditional interventions later. Does **not** by itself authorize adaptive- α . Soft pilot only.

Artifact: `artifacts/phase13-geometry-panel.json`. Protocol: `docs/PHASE13-PROTOCOL.md`.

15 Phase 14 (frozen Aug 2026 — gated steer; gate NO)

Question: can a frozen cosine gate ($\tau=-0.690$ from Ph13) decide when to apply Ph10 steer at $\alpha=4$ on **new** notes?

Result: gate **NO**. Contradiction controls and N1 stayed STEER OFF. Only 1/3 course crossed τ ; that case moved margin but did not flip X. Utility flips = 0.

Reading: soft negative — this $\tau+\alpha=4$ recipe is not a sufficient conclusion of the Ph9–13 arc. Freeze; do not retune on this held-out or jump to adaptive- α .

Artifact: `artifacts/phase14-gated-panel.json`. Protocol: `docs/PHASE14-PROTOCOL.md`.

16 What we are (and are not) claiming

- **Are:** a small synthetic lab that makes the neural→symbolic handoff *inspectable*.
- **Are not:** a clinical product, a medical device, or the full CXR stack (no Qdrant / Archetypes here).
- **Public demo:** frozen files only — safe to share; no inference on the author's servers.

17 Where to go next

Want...	Open...
Click around	https://udonsikalu.github.io/cxr-evidence-grounding-lab/notes/progress-notes.pdf
Every number & freeze	docs/PHASE9-NOTES-SIMPLE.md
Phase 9 simple notes	docs/PHASE10-PROTOCOL.md
Phase 10 (activation steer)	http://127.0.0.1:8256/ (cxr-evidence-grounding-lab-repeng/)
Rep-eng explorer	docs/PHASE11-PROTOCOL.md
Phase 11 generalization	docs/PHASE12-PROTOCOL.md
Phase 12A margin / α panel	docs/PHASE12B-PROTOCOL.md
Phase 12B second held-out	docs/PHASE13-PROTOCOL.md
Phase 13 geometry	docs/PHASE14-PROTOCOL.md
Phase 14 gated steer	docs/ARCHITECTURE-DIRECTION.md
Architecture direction	EVALUATION-JOURNEY.md
Written journey	<code>python3 server.py</code> → http://127.0.0.1:8253/
Run the local UI	GitHub project homepage
Repo README (NeSy intro)	

Rebuild: `pdflatex progress-notes-simple.tex` from the `notes/` directory.